

위상수학개론 (TP313-01)

학기	2026년 2학기
담당교수	Hudson Thomas Bryan
이수구분	교선�필수
학점	3.0
강의시간	월16:30-18:00(E7 - L22), 수16:30-18:00(E7 - L22)
강의실	E7 L12 (교수실)
소속	기초학부

교과개요

Topology, the study of “places” is a basic subject in mathematics, which studies the properties of geometric objects that are preserved by “continuous” deformations. It differs from geometry in the fact that it does not require the ability to measure distances. As topology is needed to understand in detail fundamental concepts like continuity and limits, it is a foundational subjects in modern mathematics, . While in the case of one-dimensional analysis intuition can to large extents be used to guide students to a good level of understanding, in higher dimensions knowledge of topology is essentially irreplaceable.

교과목표

The goal of this course is to introduce students to the basic notions and terminology of the subject so that it naturally fits within the modern language of category theory. Students will also be exposed to some basic aspects of algebraic topology, like the fundamental group and higher homotopy groups.

주차별 강의계획

- 1주차:Definition of topological spaces, Bases and subbases of a topology
- 2주차:Separation axioms, Metric spaces
- 3주차:Sequences and their convergence, Sequential continuity
- 4주차:Continuity, Basics of category theory
- 5주차:Examples of categories, Universal properties
- 6주차:Product and coproduct topologies, Induced topology
- 7주차:Connected, path-connected and compact topological spaces
- 8주차:Midterm exam
- 9주차:Equivalence relations and group actions, Quotient topology
- 10주차:Topological manifolds, Homotopy of paths
- 11주차:Construction of the fundamental group, The fundamental group of the circle
- 12주차:Covering spaces, Universal covering spaces
- 13주차:Seifert--Van Kampen’s theorem and applications
- 14주차:Basics of homological algebra
- 15주차:Graph theory, persistent homology and applications to Data Analysis
- 16주차:Final exam

수업방법

The course will mainly consist of frontal lectures. Homework sheets will be provided on a biweekly basis.

학습윤리

Cheating and plagiarism will not be tolerated and, if discovered, will lead to failing the entire course. This also applies to students who knowingly allow their work to be copied.

교과정책(방침)

In order to pass the course students are expected to score at least 20% on both the midterm and the final. In particular, not attending either the midterm or the final will result in a failing grade. Exceptions to this general rule will be granted on a case-by-case basis at the discretion of the instructor. Make-up exams will be granted only in exceptional cases, following the submission of appropriate documentation explaining the circumstances.